

Fish Length Estimation with Vision Foundation Models: Can They Beat the AutoFish CNN Baseline?

Abu Bakar · Laksh Jiwani · Shahman Butt

Area Seminar: Deep Learning for Maritime Vision Applications, Visual and Analytic Computing in Ocean Technologies (VACOT), University of Rostock · Supervisor: Bohan Zhuang, M.Sc. · Professor: Stefan Oehmcke

Motivation & Question

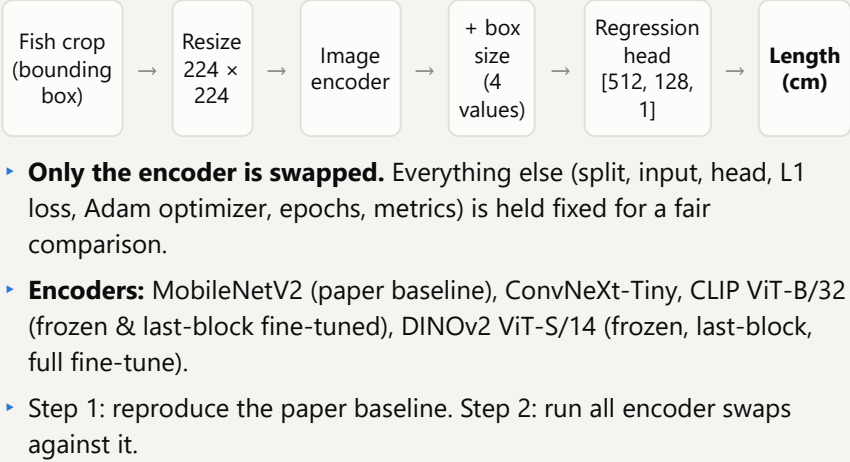
- Fish length is a key measurement for fisheries science and catch monitoring. Manual measuring is slow and invasive.
- The AutoFish benchmark (Bengtson et al., WACVW 2025) provides images, annotations, and a CNN length-regression baseline.
- Vision foundation models promise strong general features, but do they help **precise metric regression**?

Question: Do foundation-model encoders (DINOv2, CLIP) or newer supervised encoders (ConvNeXt) improve fish length estimation over the reproduced AutoFish MobileNetV2 baseline, using identical data, splits, and metrics?

Data & Setup

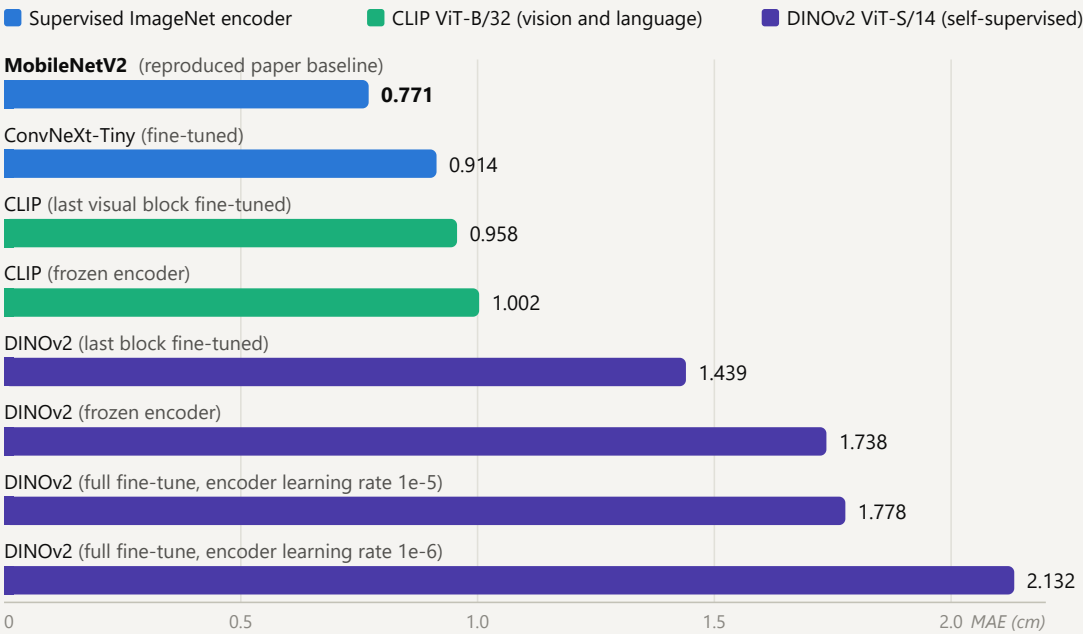
- AutoFish dataset:** 1,500 top-view images, 454 unique fish, 18,157 fish annotations, organized in 25 fish groups.
- Official group-level split:** 15 train / 5 validation / 5 test groups. No fish appears in two splits (one duplicate fish identity was found and removed).
- Two test regimes:** non-occluded (Set1+Set2, fish laid separately) vs. occluded (“All”, fish overlapping).
- Metric:** mean absolute error (MAE) of predicted length in cm, lower is better. RMSE, MAPE, and R² are tracked as secondary metrics.

Method / Workflow



Main result: the lightweight MobileNetV2 baseline still wins

Mean absolute error (cm) on the full test set (3,759 fish, occluded + non-occluded). Lower is better.



Takeaway: every foundation-model variant trails the fine-tuned MobileNetV2 baseline. Task-adapted supervised encoders (blue) fill the top of the ranking; general-purpose features transfer only partially (CLIP) or poorly (DINOv2).

Figure: full-test MAE per model, identical split / input / head / training budget for all encoders. Single training run per model (seed 42); multi-seed validation is ongoing work.

Baseline reproduction

Non-occluded test MAE (Set1+Set2), our run vs. the AutoFish paper:

Paper (Bengtson et al.)

0.62 cm

Our reproduction

0.633 cm

A difference of only 0.013 cm, so the baseline and our whole comparison pipeline can be trusted.

Key findings

- Baseline reproduced** within 0.013 cm of the paper on non-occluded fish.
- MobileNetV2 remains best overall:** 0.771 cm full-test MAE.
- Foundation features are not automatically better.** Frozen DINOv2 reaches only 1.738 cm; even its best variant (last-block fine-tuned) stays at 1.439 cm.
- CLIP transfers far better than DINOv2** (0.958 cm with light fine-tuning) but still does not beat the CNN.
- Supervised ImageNet encoders are the closest challengers:** ConvNeXt-Tiny reaches 0.914 cm.
- Occlusion hurts every model:** e.g. the baseline degrades from 0.633 cm (separate fish) to 0.909 cm (overlapping fish).

Take-home message

For precise fish length regression, a small task-adapted convolutional network still beats vision foundation models. Strong general features are no substitute for end-to-end task adaptation when millimetres matter.

Limitations

- One training run per model (fixed seed), so differences of a few hundredths of a cm are not yet statistically validated.
- Foundation models were tested at small scale (ViT-S/14, ViT-B/32) with limited fine-tuning budgets. Larger variants might behave differently.
- No per-species or per-length-range error analysis yet.

Next steps

- EfficientNet-B0 as the next supervised-encoder candidate.
- Repeat the top models with 3 seeds to quantify run-to-run variance.
- Error analysis by species, fish size, and occlusion difficulty.